Facilitating the Day 7 IPC role play

Trainer's crib sheet · Module 6, Day 7 session 4 · 15:30–17:00

Groups of four — genomics lead, IPC nurse, neonatologist, microbiology lab manager — present the ST39 findings to a mock IPC panel: 5 minutes to present, 10 for questions, then two agreed actions. Below are model answers for the four scripted questions, each paired with the weak answer groups usually give. The contrast is the teaching point; read the weak answers before the session so you recognise them live.

The session is about **translating evidence into a decision under uncertainty**, not about recalling tool output. A group that gets every SNP number right and recommends nothing has not passed.

The data the groups are working from

Isolate(s)	Ward	Year	Role in the exercise
Cluster (8 `CAA*`)	Chatinkha Nursery	2017	The putative outbreak
`BKR8JH`	Chatinkha Nursery	2016	**The hinge**: reservoir vs new introduction
`D29438`	unknown	2005	Background ST39
`BHB7VE`	4A Female Medical Ward	2016	Different ward; carries NDM-1
`CHI11E`	unknown	2019	Hybrid isolate; shares a plasmid with ST45 `CNS1F1`

From earlier days: CTX-M-15 in 21 of 29 isolates; acquired gentamicin resistance genes in most; root-to-tip regression on the ST39 group shows **no clear temporal signal**, so the cluster cannot be dated.

1. “So is it an outbreak or not?”

STRONG

“Eight isolates from Chatinkha Nursery in 2017 are all ST39 and within the SNP threshold of each other. That is consistent with **probable transmission within the neonatal unit**. Whether you call it an outbreak is your call, not mine — that depends on your case definition and baseline rate, which genomics does not supply. What I can say is that these eight are not independent introductions.”

WEAK

“Yes, it's an outbreak” (not the genomics lead's designation to make), or “the SNPs prove it” (they support a hypothesis; they rarely settle it).

Credit any group that volunteers the **Snippy vs SKA2 cross-check** without prompting: a pair that is close by one method and not the other is at most *possible* transmission until they know why.

2. “Did it come from one baby, or from the unit itself?”

The reservoir-versus-point-source question. It turns on `BKR8JH`.

STRONG

“If `BKR8JH` sits within a few SNPs of the 2017 cluster, this lineage was already in the ward in 2016, which points to a **persistent reservoir** — sinks, incubators, shared equipment, colonised babies or staff — rather than one index baby. If it is far away, the 2017 cluster is more likely a recent introduction that then spread between babies. I can tell you which, but I **cannot name an index case**: all eight are from the same year and the temporal signal is too weak to date the common ancestor or order the transmissions.”

WEAK

Naming a specific baby as the source, or claiming the tree shows the direction of transmission.

Credit two further observations:

- **Tree shape.** Eight isolates in a tight polytomy with near-zero distances fits a common source or rapid spread; a ladder-like topology fits sequential baby-to-baby transmission. With few SNPs the topology is weakly supported, so bootstrap values should be checked before leaning on it.
- **No environmental or screening samples were taken**, so a reservoir can be neither confirmed nor excluded — which is what motivates the first action below.

3. “What should we do on Monday morning?”

Actions must be specific and testable. Two good choices, with the reasoning that earns the marks:

> **STRONG — ACTION 1**
>
> **Environmental sampling in Chatinkha Nursery** — sinks and drains, incubator water reservoirs, shared equipment; culture on ESBL-selective media, sequence whatever grows, compare with the cluster. The one action that can confirm or exclude the reservoir hypothesis `BKR8JH` raises.

> **STRONG — ACTION 2**
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> **Admission and point-prevalence screening** (rectal swabs) over a defined period. This answers the question IPC will ask next — *is transmission still happening?* — which 2017 isolates cannot.

> **WEAK**
>
> “More training” or “better hand hygiene” with no audit, no denominator and no way to tell whether it worked. Push for: **what will you measure, and when will you know?**

Also defensible: a hand-hygiene and cleaning audit using **observed** compliance rather than self-report; a review of vascular access practice, since these are bloodstream infections; prospective sequencing of new *Klebsiella* blood isolates from the unit.

4. “Will treatment change?”

> **STRONG**
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> “For these babies, no — treatment follows their AST, which your lab already has. For the unit's **empiric guideline** there is a problem worth reviewing. These isolates carry CTX-M-15 and gentamicin resistance genes. Ampicillin is intrinsically ineffective in *K. pneumoniae*, so ampicillin + gentamicin rests on gentamicin alone — and that is predicted to fail. CTX-M-15 removes third-generation cephalosporins too. Both first and second line are compromised in this lineage, which leaves a carbapenem.”

Three caveats separate a good answer from a reckless one. Challenge a group that misses all three:

- **This is not a new guideline.** Twenty-nine hand-picked isolates that over-represent clusters cannot give a failure rate. Changing empiric therapy needs the lab's full AST over a defined period, with a denominator, plus stewardship and paediatric input.
- **Carbapenem resistance is already in the hospital.** `BHB7VE`, from 4A Female Medical Ward, carries NDM-1. Broad carbapenem use in the nursery selects for exactly that.
- **Amikacin must be checked against AST**, not assumed — the Day 4 major-error discussion, where *aac(6')-Ib-cr* often fails to cross the breakpoint.

An extra card for the lab manager

`CHI11E`'s ~143 kb IncFII(K)/IncR plasmid is about 99.95% identical to the one in `CNS1F1`, a **different ST**. The resistance itself moves between lineages. Useful when the panel asks “if we stop this clone, is the problem solved?” — the answer is no, and it argues for watching resistance genes and plasmids, not only STs.

Marking the session

Credit groups that volunteer the limitations before being asked: only cultured and sequenced isolates are visible (colonised babies are not); no environmental or screening samples; year-only dates, so events cannot be ordered; a selected training set; SNP thresholds that are guides rather than rules.

> **WATCH FOR**
>
> The failure mode is not only overclaiming. A genomics lead who says “we cannot be certain” four times and recommends nothing has also failed. The target is: **state what the data support, state what they do not, and still name two actions.**

Practicalities. 15 minutes per group, so with four groups allow an hour plus 15 minutes to agree the actions as a whole room. If you are short of time, run two groups in front of everyone and have the rest critique — that works better than rushing all four.

Session notes

One row per group. Tick what they did unprompted; the last column is for the debrief.

Group	Stated the limitations	Named two testable actions	Avoided naming an index case	Careful on treatment	Notes for the debrief
1					
2					
3					
4					

Two actions the room agreed: